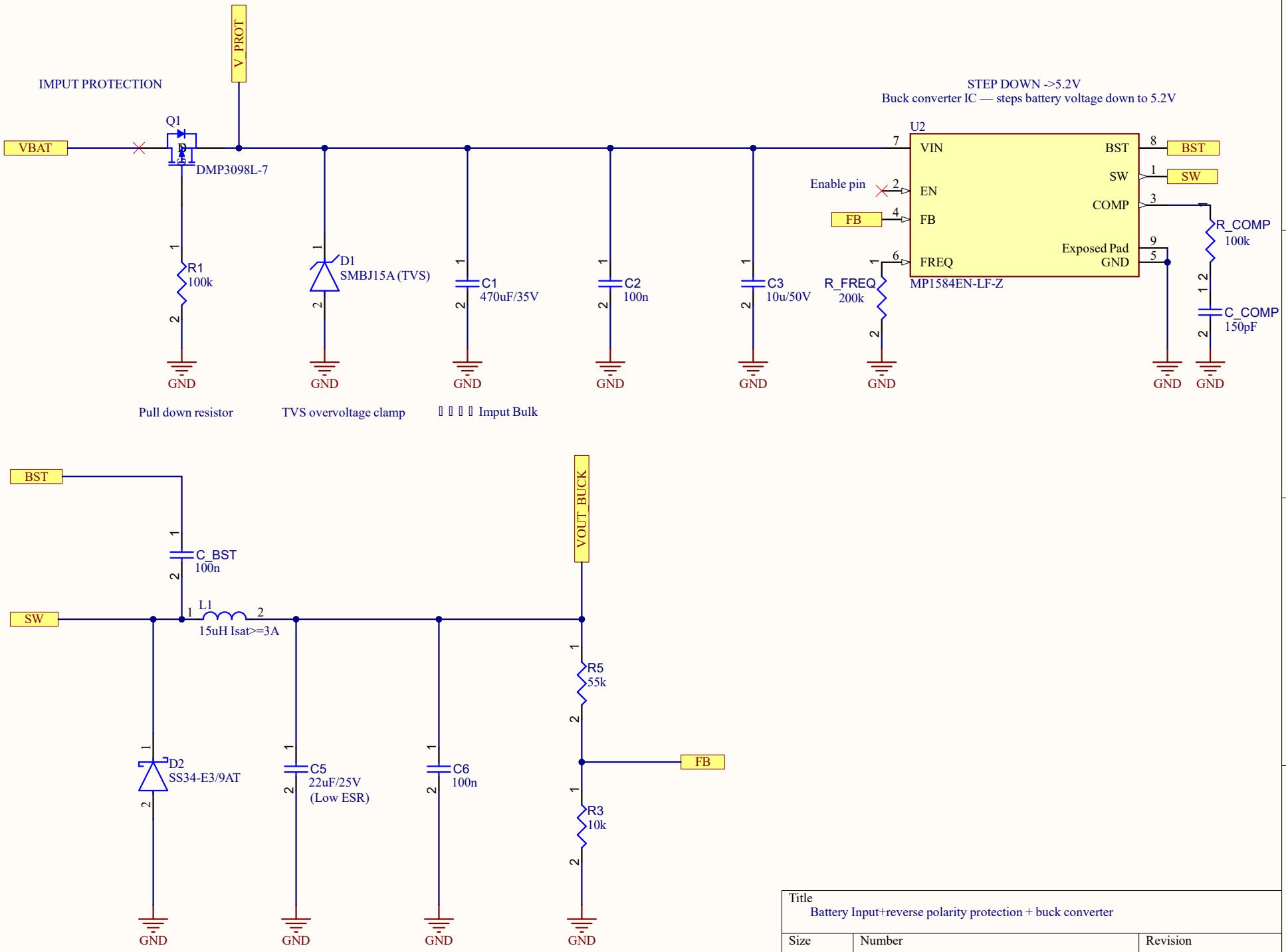
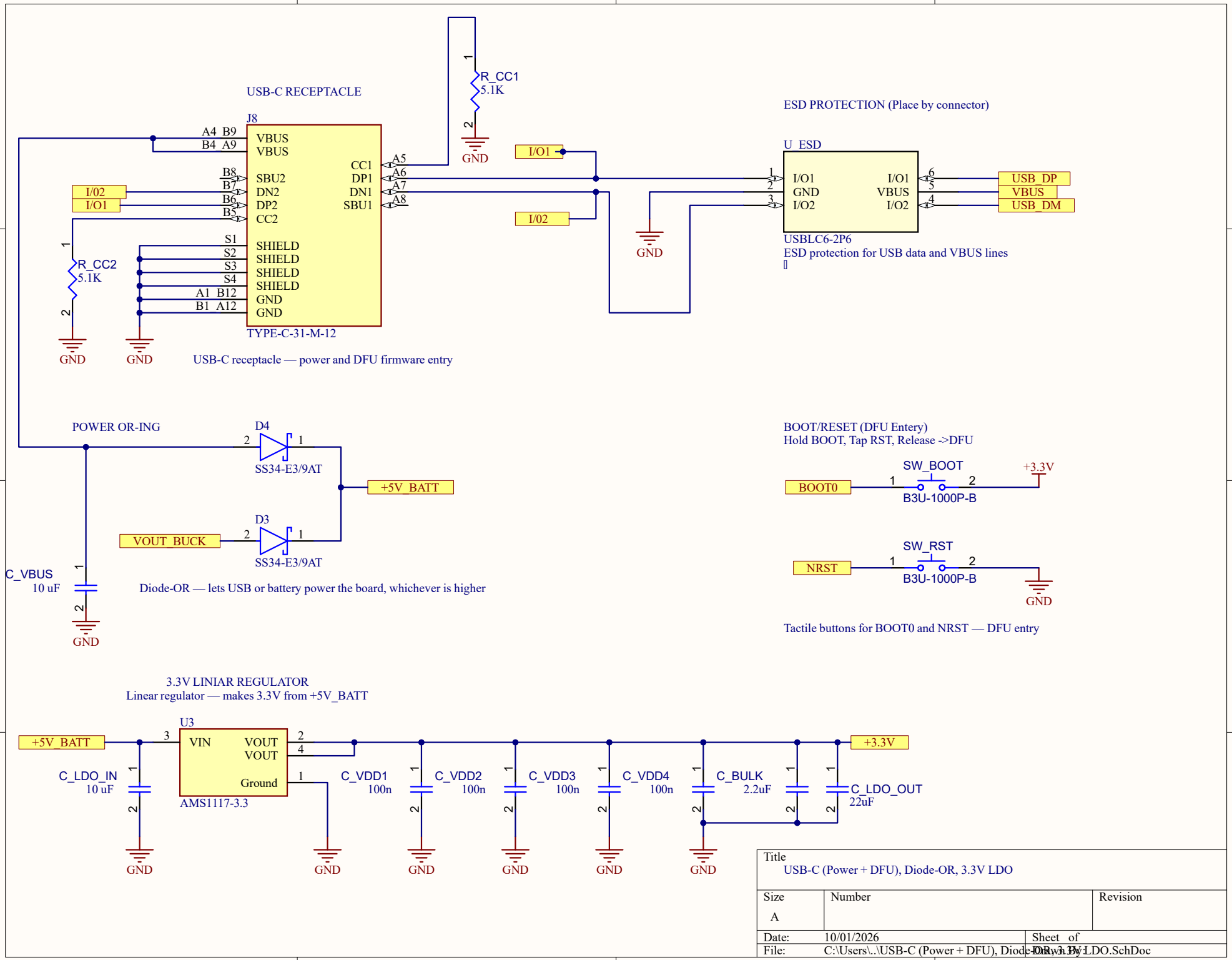
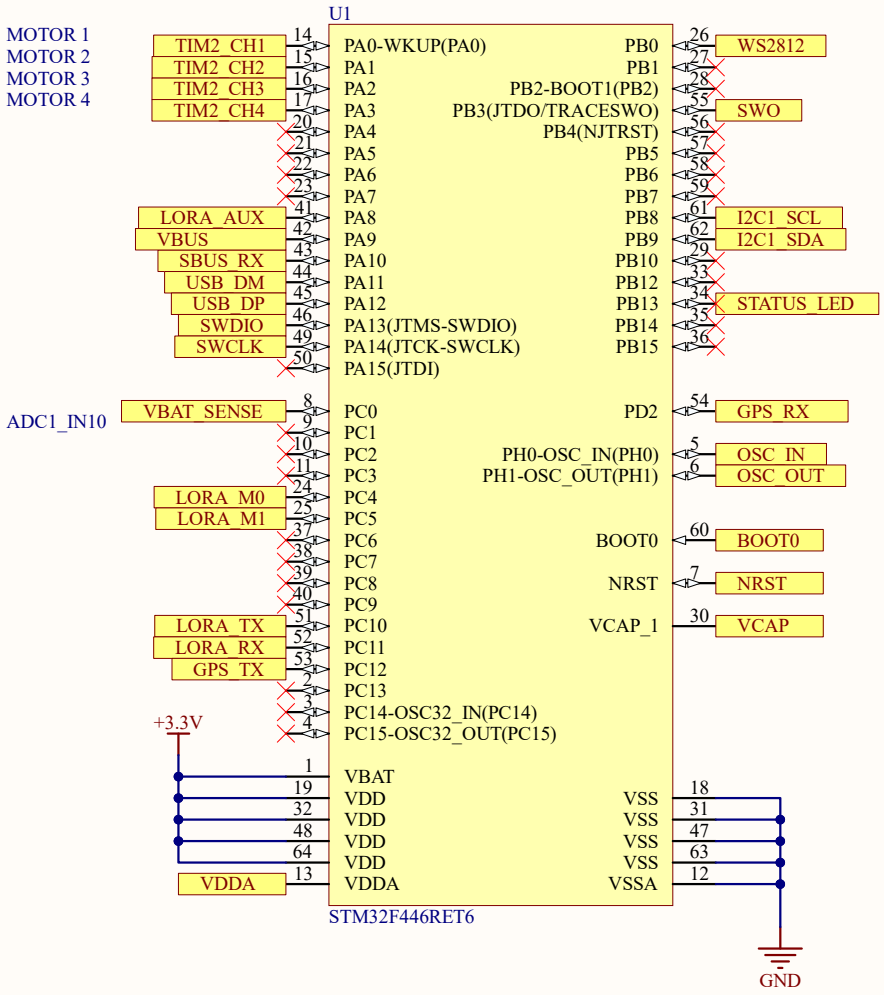
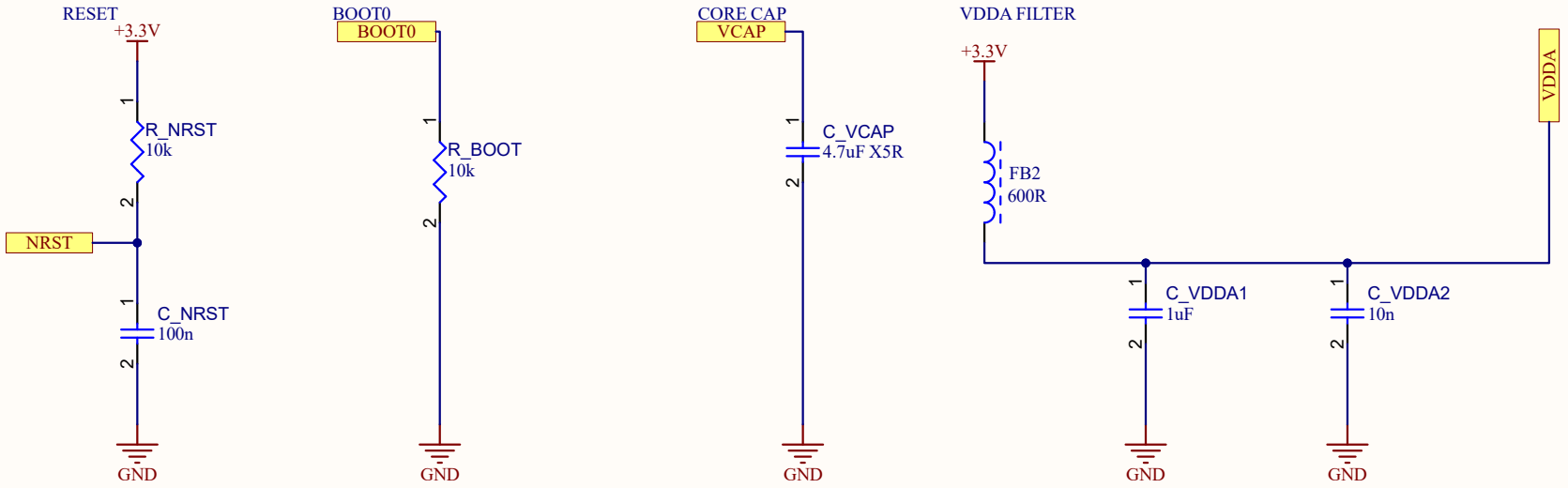

IMPUT PROTECTION
V_PROT
VBAT
Q1
DMP3098L-7
R1
100k
D1
SMBJ15A (TVS)
C1
470uF/35V
C2
100n
C3
10u/50V
GND
Pull down resistor
TVS overvoltage clamp
Imput Bulk
STEP DOWN ->5.2V
Buck converter IC — steps battery voltage down to 5.2V
U2
VIN
BST
SW
COMP
EN
FB
FREQ
Exposed Pad
GND
MP1584EN-LF-Z
Enable pin
FB
R_FREQ
200k
R_COMP
100k
C_COMP
150pF
BST
SW
C_BST
100n
L1
15uH Isat>=3A
VOUT_BUCK
D2
SS34-E3/9AT
C5
22uF/25V
(Low ESR)
C6
100n
R5
55k
R3
10k
FB
Title
Battery Input+reverse polarity protection + buck converter
Size
A
Number
Revision
Date:
10/01/2026
Sheet of
File:
C:\Users\..\Battery Input, Protection, 5.2V Buck.SchDoc
Drawn By:


USB-C RECEPTACLE
J8
VBUS
VBUS
SBU2
DN2
DP2
CC2
SHIELD
SHIELD
SHIELD
SHIELD
GND
GND
CC1
DP1
DN1
SBU1
TYPE-C-31-M-12
R_CC1
5.1K
R_CC2
5.1K
I/02
I/O1
GND
USB-C receptacle — power and DFU firmware entry
ESD PROTECTION (Place by connector)
U_ESD
I/O1
GND
I/O2
VBUS
USB_DP
USB_DM
USBLC6-2P6
ESD protection for USB data and VBUS lines
POWER OR-ING
D4
SS34-E3/9AT
D3
SS34-E3/9AT
+5V_BATT
VOUT_BUCK
C_VBUS
10 uF
Diode-OR — lets USB or battery power the board, whichever is higher
BOOT/RESET (DFU Entery)
Hold BOOT, Tap RST, Release ->DFU
SW_BOOT
BOOT0
+3.3V
B3U-1000P-B
SW_RST
NRST
B3U-1000P-B
Tactile buttons for BOOT0 and NRST — DFU entry
3.3V LINIAR REGULATOR
Linear regulator — makes 3.3V from +5V_BATT
U3
+5V_BATT
VIN
VOUT
VOUT
Ground
AMS1117-3.3
C_LDO_IN
10 uF
C_VDD1
100n
C_VDD2
100n
C_VDD3
100n
C_VDD4
100n
C_BULK
2.2uF
C_LDO_OUT
22uF
+3.3V
Title
USB-C (Power + DFU), Diode-OR, 3.3V LDO
Size
A
Number
Revision
Date:
10/01/2026
Sheet of
File:
C:\Users\..\USB-C (Power + DFU), Diode-OR, 3.3V LDO.SchDoc
Drawn By:



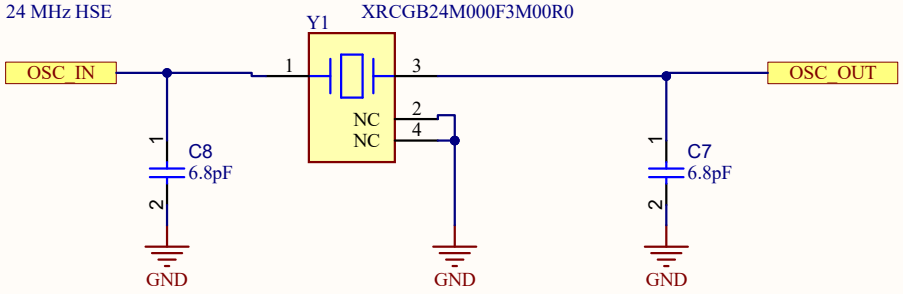
Title: STM32F446RET6 Core		
Size: A	Number:	Revision:
Date: 10/01/2026		Sheet of
File: C:\Users\..\STM32F446RET6 Core.SchDoc		Drawn By:

Reset pull-up and debounce for NRST



Pull-down for BOOT0 — normal boot from flash by default

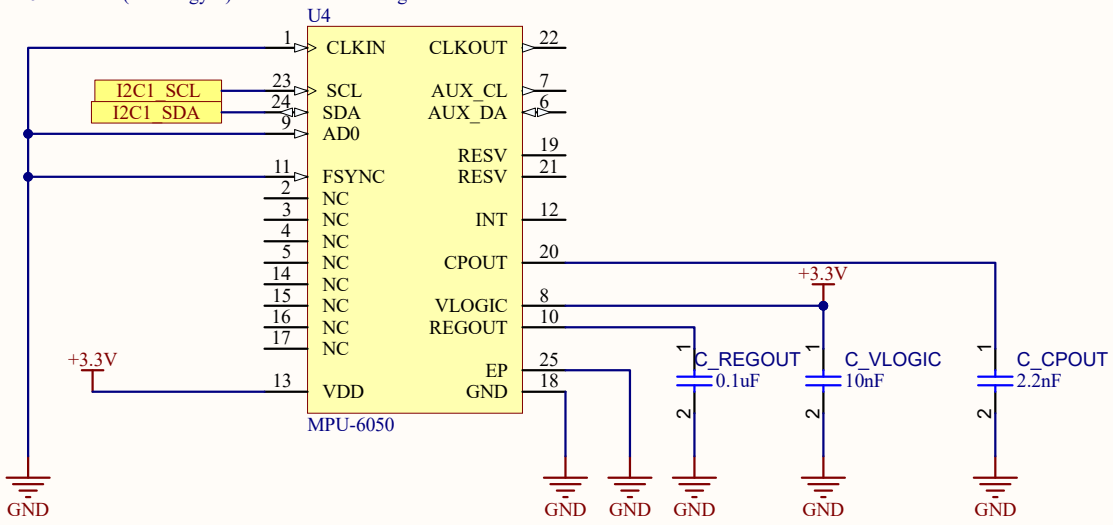
24 MHz HSE



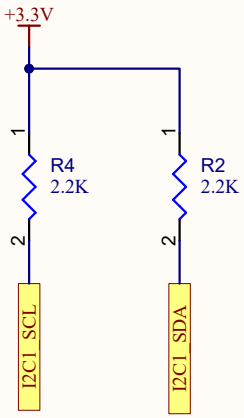
Main clock source for the MCU

Title: MCU Support Circuits		
Size: A	Number:	Revision:
Date: 10/01/2026		Sheet of
File: C:\Users\..\MCU Support Circuit.SchDoc		Drawn By:

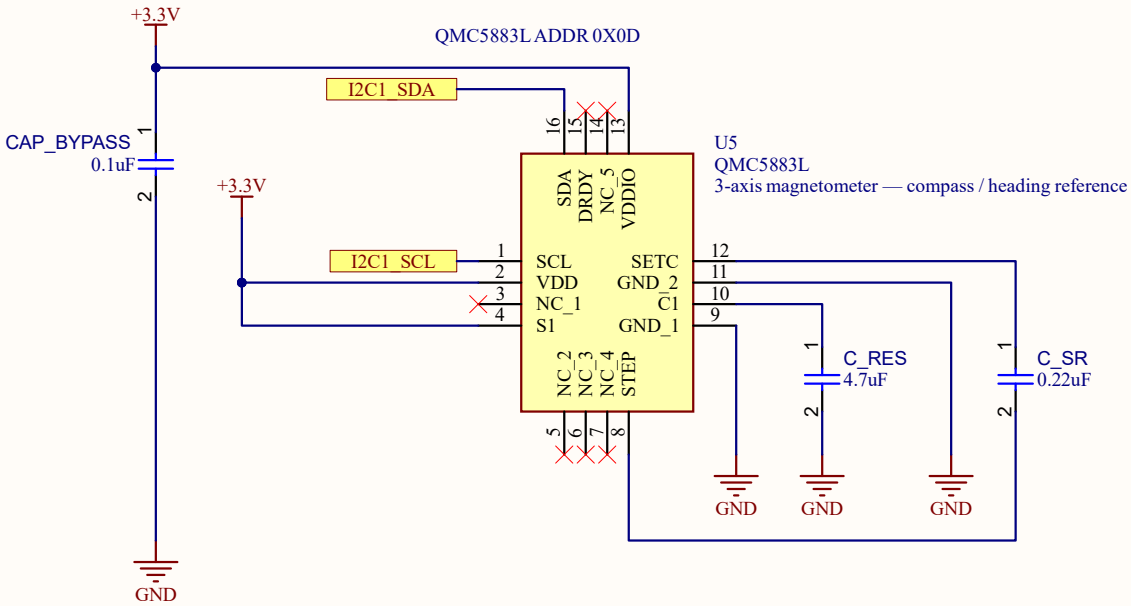
MPU-6050 ADDR 0X68
6-axis IMU (accel + gyro) — core sensor for flight stabilization



BUS PULL-UPS

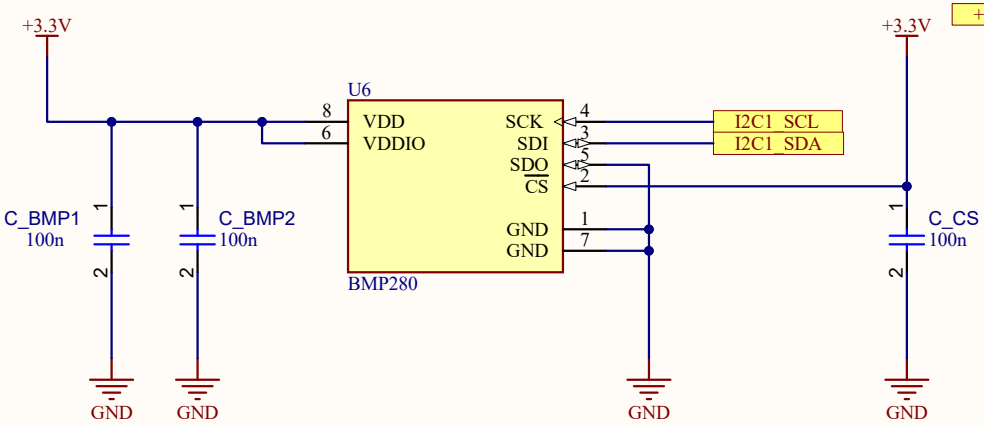


QMC5883L ADDR 0X0D

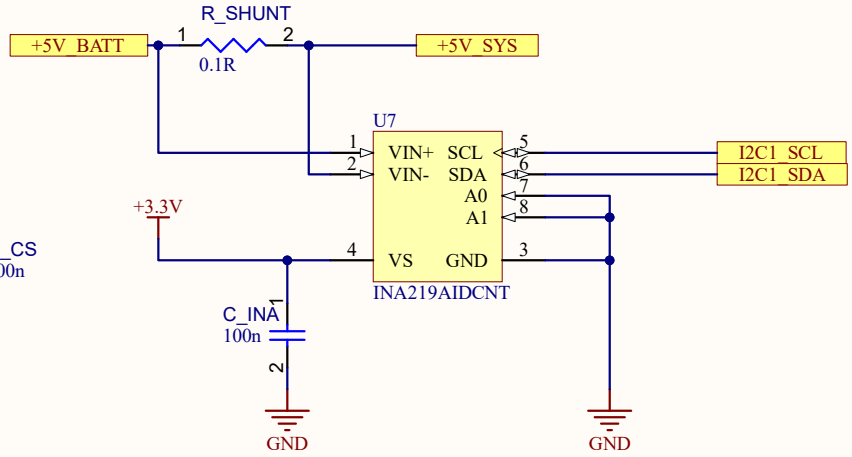


Title: I2C Sensors A- IMU and Magnetometer		
Size: A	Number:	Revision:
Date: 10/01/2026		Sheet of
File: C:\Users\..\I2C Sensors A- IMU and Magnetometer.SchDoc		Drawn By:

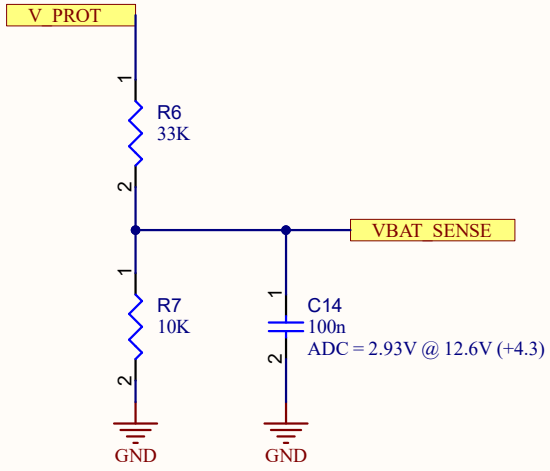
BMP 280 ADDR 0X76
Barometric pressure sensor — used for altitude



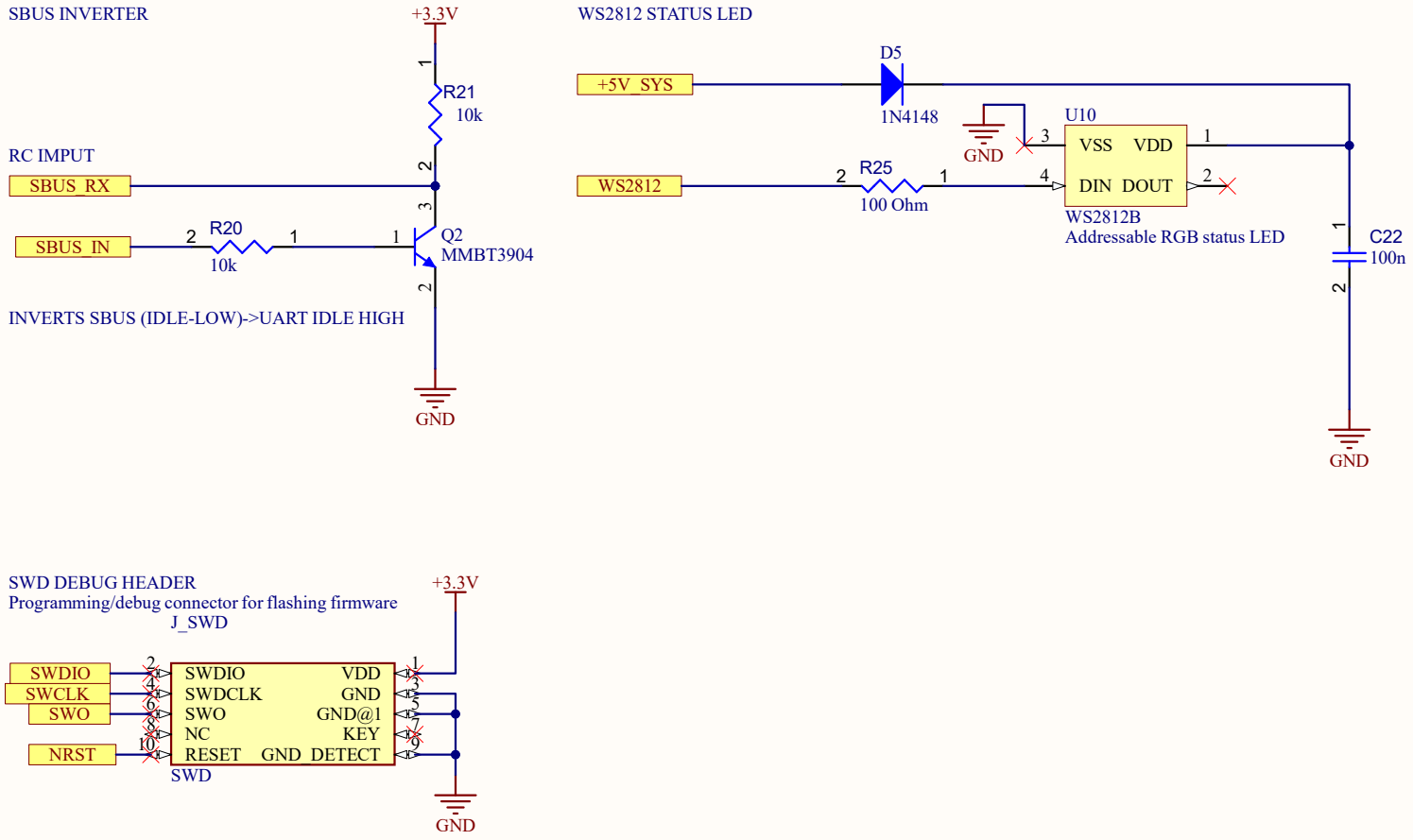
INA219 + SHUNTADDR 0X40



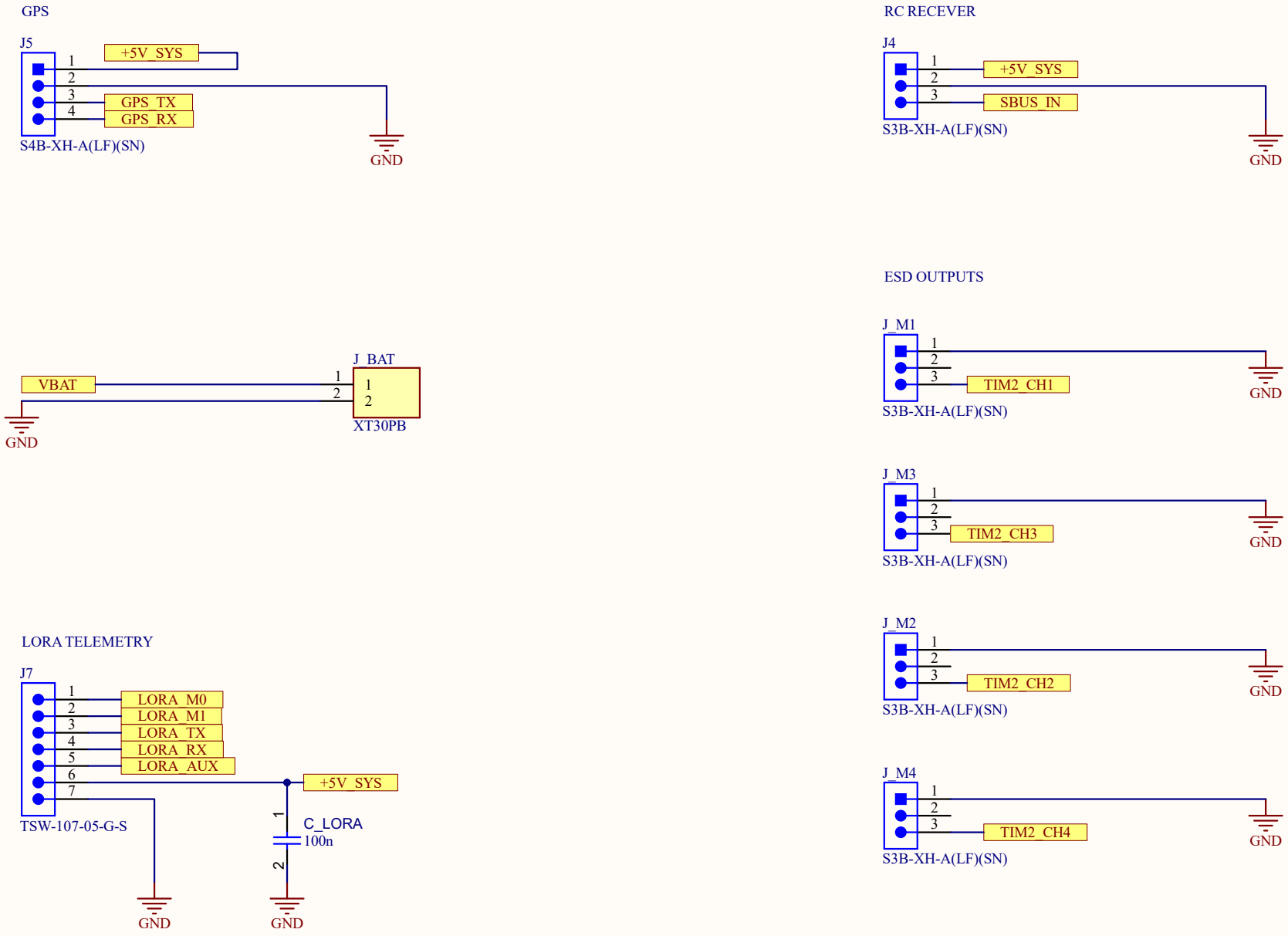
BATTERY SENSE



Title: I2C SENSORS B - BAROMETER, CURRENT MONITOR, BATTERY SENSE		
Size: A	Number:	Revision:
Date: 10/01/2026		Sheet of
File: C:\Users\..\I2C SENSORS B - BAROMETER, CURRENT MONITOR, BATTERY SENSE.		Drawn By:



Title: RC INPUT, STATUS LED, SWD HEADER		
Size: A	Number:	Revision:
Date: 10/01/2026		Sheet of
File: C:\Users\..\RC INPUT, STATUS LED, SWD HEADER.sch		Drawn By:


GPS
J5
+5V_SYS
GPS_TX
GPS_RX
S4B-XH-A(LF)(SN)
GND
RC RECEVER
J4
+5V_SYS
SBUS_IN
S3B-XH-A(LF)(SN)
GND
ESD OUTPUTS
J_M1
TIM2_CH1
S3B-XH-A(LF)(SN)
GND
J_M3
TIM2_CH3
S3B-XH-A(LF)(SN)
GND
J_M2
TIM2_CH2
S3B-XH-A(LF)(SN)
GND
J_M4
TIM2_CH4
S3B-XH-A(LF)(SN)
GND
VBAT
J_BAT
XT30PB
GND
LORA TELEMETRY
J7
LORA_M0
LORA_M1
LORA_TX
LORA_RX
LORA_AUX
+5V_SYS
TSW-107-05-G-S
C_LORA
100n
GND
GND
Title
EXTERNAL CONNECTORS
Size
A
Number
Revision
Date:
10/01/2026
Sheet of
File:
C:\Users\..\EXTERNAL CONNECTORS.SchDoc
Drawn By: